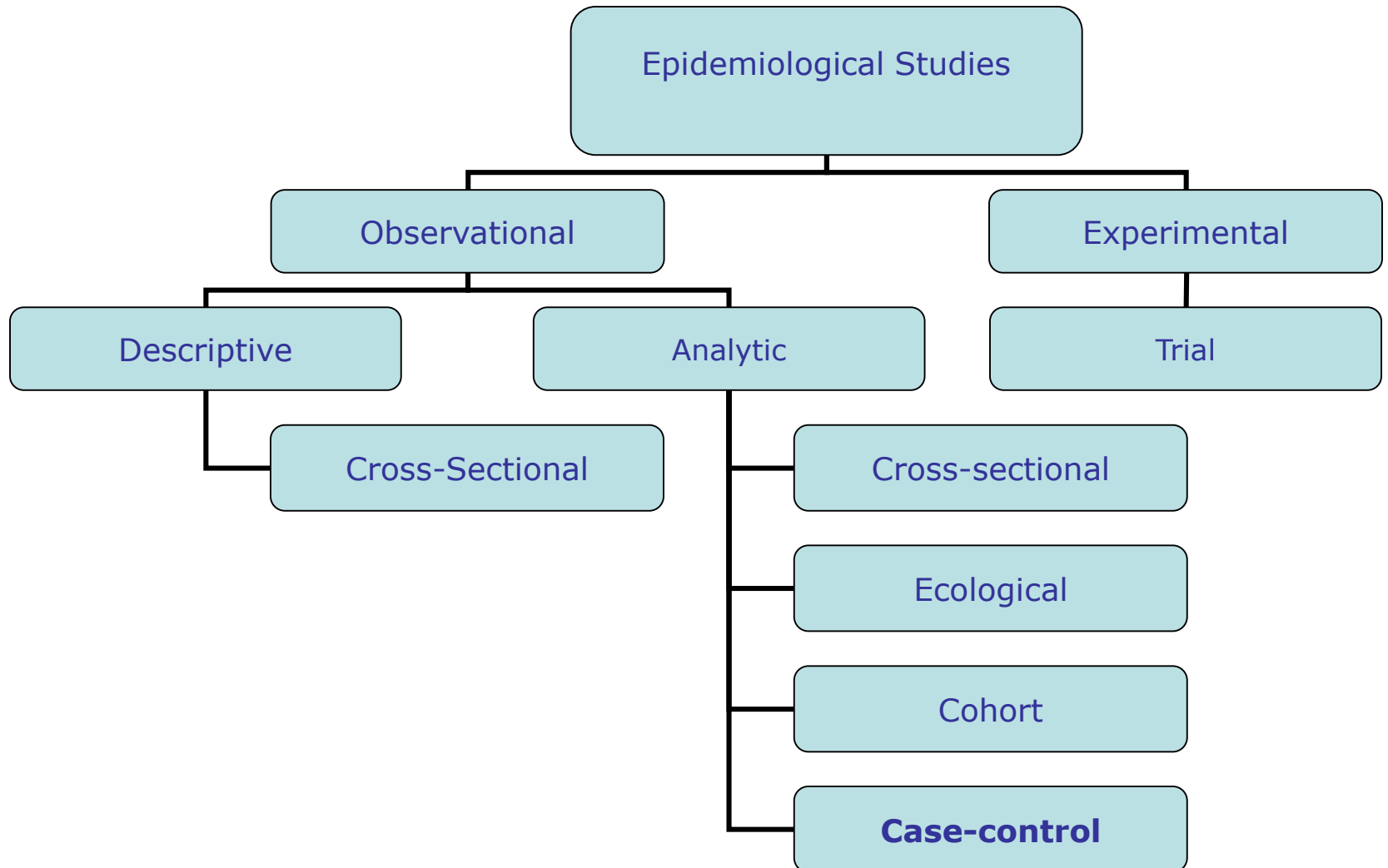


Study Design II

The 'tree' of study designs



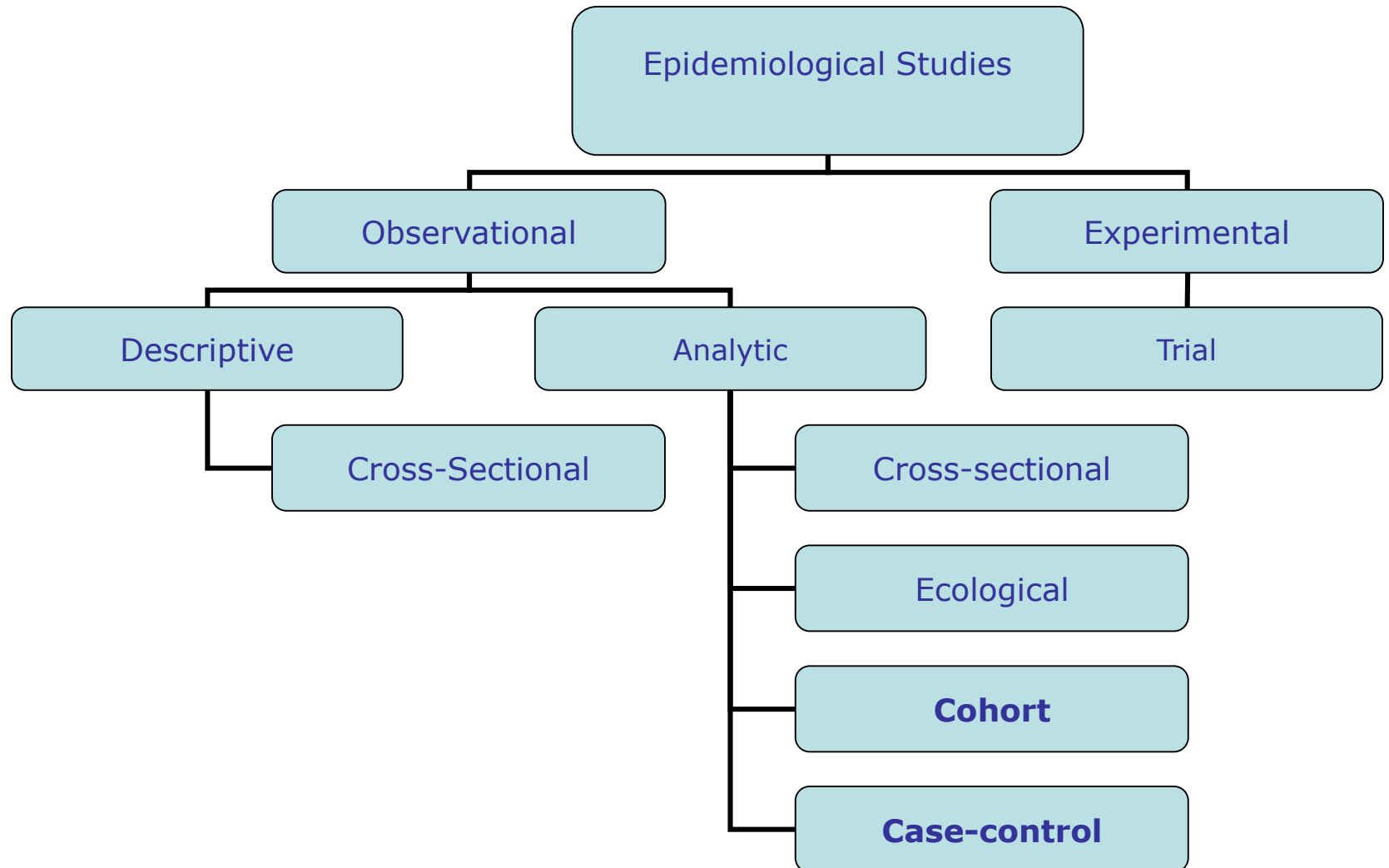
Analytic Study:

Observational studies, where the primary goal of a study is establishing a relationship (association) between a 'risk factor' (etiological agent) and an outcome (disease), it is termed analytical. Analytical studies always require having a comparison group.

There are two types

- Case Control Study
- Cohort Study

The 'tree' of study designs



- The main advantage of an analytic study is the possibility of analysis: measuring association.

Measures of Association

- In epidemiology the calculation of appropriate measures of disease frequency is the basis for the comparison of populations and identification of disease determinants
- Measuring the occurrence of disease or other health states is only the beginning of epidemiological process and the next step is the comparison of occurrence in two or more groups whose exposure have differed

Exposed group

Unexposed group

Examples

- Is there association between OCP use and development of heart problems?
- Is there association between seat-belt use and mortality (death) from car accident?
- Is there association between the educational status of mother and the nutritional status of their children?

Measures of Association

The two frequencies being compared can be combined into a single summary parameter that estimates the association between an exposure and risk of developing a disease by calculating either the *ratio* of the measures of disease frequency for the two populations or the *difference* between the two

Measures of Association

To aid calculation of measures of association, epidemiological data are often presented in the form of *two-by-two tables* (four fold/contingency table)

Contingency Table

Disease

Yes

No

Total

Exposure

Yes

a

b

$a+b$

No

c

d

$c+d$

Total

$a+c$

$b+d$

$a+b+c+d$

Example 1 (Case-control study)

		Myocardial Infarction		Total
		Yes	No	
OC use	Yes	23	304	327
	No	133	2816	2949
Total		156	3120	3276

Example 2 (Cohort study)

		Bacteruria		Total
		Yes	No	
OC Use	Yes	27	455	482
	No	77	1831	1908
Total		104	2286	2390

Case control study

- Case-control study design is design where by people diagnosed as having a disease (cases) are compared with persons who do not have the disease (controls) to determine if the two groups differ in the proportion of persons exposed to a specific factor or factors.

- **Case-control** study design is the most commonly used analytical strategy in epidemiology. The design is more appropriate in the clinical setting, because the hall-mark of a case-control study is that it begins with the disease (**cases**) and compares them with people without the disease (**controls**).
- **The case-control** design is relatively simple; except that it is backward-looking (**retrospective**) based on the exposure histories of cases and controls. With this type of study, comparison is made between individuals with disease of interest and individuals free of the disease of interest.

- Data are analyzed to determine whether exposure was different for cases and for controls. The risk factor is something that happened or began in the past, presumably before disease onset.
- Information about the exposure is obtained by taking a history and/or from records.
- Occasionally, the suspected factor or attribute is a permanent one, such as blood group, which can be ascertained by clinical or laboratory investigation.

Past

Present

Retrospective
study

Look for past
exposure to factor in
cases and control

Select cases and
controls



Selection of cases

- What constitutes a case in the study should be **clearly defined** with regard to specific characteristics of the disease. Cases that do not fit these criteria should be excluded from the study.
- This design is particularly efficient for rare diseases, because all cases that fit the study criteria in a particular setting within a specific period are usually included.
- This allows for a reasonable number of cases to be included in the study without waiting for the occurrence of new cases of the disease, which might take a long time.

Selection of controls

- It is crucial to set up one or more control groups of people who do not have the specified disease or condition in order to obtain estimates of the frequency of the attribute or risk factor for comparison with its frequency among cases.
- This is the most important aspect of the case-control study, as biases in the selection of controls may invalidate the study results, and bias in the selection of controls is often the greatest cause for concern when analyzing data from case-control studies.

The sources of control groups may be:

- a probability sample of a defined population, if the cases are drawn from that defined population;
- a sample of patients admitted to, or attending the same institution as the cases;
- a sample of relatives or associates of the cases (neighborhood controls);
- a group of persons selected from the same source population as the cases, and matched with the cases for potentially confounding variables; on other risk factors (other than the one under consideration);

Advantages

- feasible when the disease being studied occurs only rarely, e.g. cancer of a specific organ;
- relatively efficient, requiring a smaller sample than a cohort study;
- little problem with attrition,
- sometimes they are the earliest practical observational strategy for determining an association.
- can examine multiple etiologic factors for a single disease.

Disadvantages

- the absence of epidemiological denominators (population at risk) makes the calculation of incidence rates, and hence of attributable risks, impossible;
- temporality is a serious problem in many case-control studies where it is not possible to determine whether the attribute led to the disease/condition, or vice versa;
- particularly prone to bias compared with other analytic designs, in particular selection and recall bias, " inefficient for the evaluation of rare exposures.

An indigenous malaria transmission in the outskirts of Addis Ababa, Akaki Town and its environs

Aduagna Woyessa¹, Teshome Gebre-Micheal², Ahmed Ali³

Abstract

Background: In recent years malaria is becoming endemic in highland areas beyond its previously known upper limit of transmission. Assessment of the situation of the disease in such areas is necessary in order to institute appropriate control activities.

Objectives: The objectives of the study were to determine the prevalence of malaria, the parasite species involved and *Anopheles* species responsible in local malaria transmission.

Methods: A systematic sampling technique was used to select survey households. Blood films were collected monthly between October and December 1999 from all household members by a trained and experienced laboratory technician. Larval and adult mosquitoes were monthly collected using different methods from September 1999 to October 2000.

Results: Among 2136 examined blood films, 78(3.7%) of them were malaria positive of which 54(69%) were due to *Plasmodium vivax* and 24 (31%) due to *P. falciparum*. *Anopheles gambiae* s. l. (presumably *An. arabiensis*) and *An. christyi* were the dominant man-biting species, with the former being the major vector in the area. Both these species were found to be more of exophagic and active in the early evening, unlike *An. pharoensis*, which showed an endophagic tendency.

Conclusion: This study indicated that indigenous transmission of malaria occurs in the study area. Transmission is reckoned to be maintained by low density of vector species for short period of time under favorable conditions. Therefore, the acquisition of communal immunity is interrupted by long duration of non-malaria season leading to the occurrence of recurrent malaria epidemics. [*Ethiop. J. Health Dev.* 2004;18(1):2-7]

Cohort Study

- A cohort study is an observational research design which begins when a group of people (a cohort) initially free of disease (outcome of interest), are classified according to a given exposure, and then followed up over time.
- The researcher compares whether the subsequent development of any new cases of a disease (or other outcome of interest) differs between the exposed and non-exposed groups.

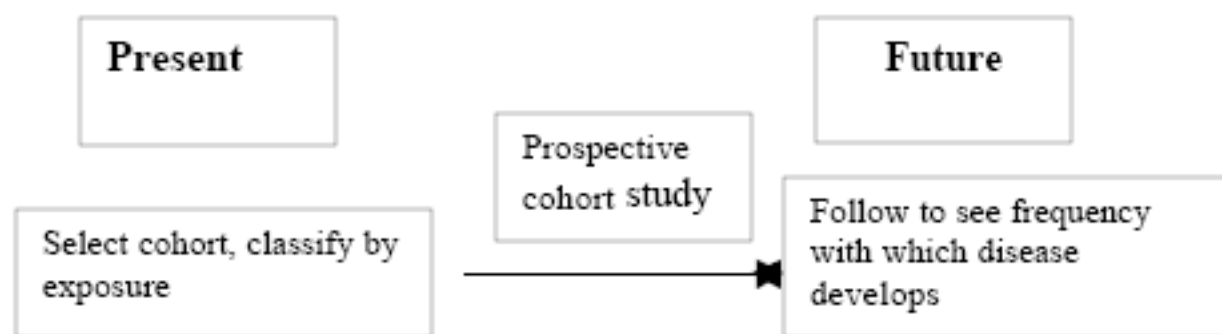


Figure 2. Schematic diagram of time factor in cohort study

Types of cohort studies

- There are basically two types of cohort studies; prospective and retrospective (historical). The difference between the two lies in where the starting point of the study is deemed to begin. In a prospective cohort study the starting point of observation (time zero) is 'now', and the population is followed into the future.
- The exposure of interest may or may not have occurred when the study was initiated, but the outcome has not.

- A historical cohort study depends upon the availability of data or records that allow reconstruction of the exposure of cohorts to a suspected risk factor and follow-up of their mortality or morbidity over time. In other words, although the investigator was not present when the exposure was first identified, he reconstructs exposed and unexposed populations from records, and then proceeds as though he had been present throughout the study.
- Historically constructed cohorts share several advantages of the prospective cohort. If all requirements are satisfied, a historical cohort may suffer less from the disadvantages of time and expense.

Disadvantages

- All of the relevant variables may not be available in the original records.
- It may be difficult to ascertain that the study population was free from the condition at the start of the comparison. This problem does not exist if we are concerned with deaths as indicators of disease.
- " Attrition problems may be serious due to loss of records, incomplete records, or difficulties in tracing or locating all of the original population for further study.
- " These studies require ingenuity in identifying suitable populations and in obtaining reliable information concerning exposure and other relevant factors.
- Examples of such population groups include military personnel, industrial groups (such as miners), professional groups, etc.

Advantages

- Relative risk can be calculated.
- Allows concluding a cause-effect relationship.
- No chance of bias being introduced due to awareness of being sick as in encountered in case-control studies.
- Less chance for the problem of selective survival or selective recall
- Cohort studies are capable of identifying other diseases that may be related to the same risk factor.
- Allows estimating attributable risks, thus indicating the absolute magnitude of disease attributable to the risk factor.
- If a probability sample is taken from the reference population, it is possible to generalize from the sample to the reference population with a known degree of precision.

Disadvantages

- Cohort studies are long-term and are thus not always feasible; they are relatively inefficient for studying rare conditions.
- Costly in time, personnel, space and patient follow-up.
- Sample sizes required are large, especially for infrequent conditions.
- Attrition or loss of people from the sample or control during the study is the major problem.
- The higher the proportion lost (say beyond 10-15%) the more serious the potential bias.
- There may also be attrition among investigators.
- Over a long period, many changes may occur in the environment, among individuals or in the type of intervention, and these may confuse the issue of association and attributable risk.
- Over a long period, study procedures may influence the behavior of the persons investigated in such a way that the development of the disease may be influenced accordingly (Hawthorne effect).
- A serious ethical problem may arise when it becomes apparent that the exposed population is manifesting significant disease excess before the follow-up period is completed.

Alcohol consumption and mortality from all causes, coronary heart disease, and stroke: results from a prospective cohort study of Scottish men with 21 years of follow up

Carole L Hart, George Davey Smith, David J Hole, Victor M Hawthorne

Abstract

Objectives: To relate alcohol consumption to mortality.

Design: Prospective cohort study.

Setting: 27 workplaces in the west of Scotland.

Participants: 5766 men aged 35-64 when screened in 1970-3 who answered questions on their usual weekly alcohol consumption.

Main outcome measures: Mortality from all causes, coronary heart disease, stroke, and alcohol related causes over 21 years of follow up related to units of alcohol consumed per week.

Results: Risk for all cause mortality was similar for non-drinkers and men drinking up to 14 units a week. Mortality risk then showed a graded association with alcohol consumption (relative rate compared with non-drinkers 1.34 (95% confidence interval 1.14 to 1.58) for 15-21 units a week, 1.49 (1.27 to 1.75) for 22-34 units, 1.74 (1.47 to 2.06) for 35 or more units). Adjustment for risk factors attenuated the increased relative risks, but they remained significantly above 1 for men drinking 22 or more units a week. There was no strong relation between alcohol consumption and mortality from coronary heart disease after adjustment. A strong positive relation was seen between alcohol consumption and risk of mortality from stroke, with men drinking 35 or more units having double the risk of non-drinkers, even after adjustment.

Conclusions The overall association between alcohol consumption and mortality is unfavorable for men drinking over 22 units a week, and there is no clear evidence of any protective effect for men drinking less than this.